

### 0.0.1 Field Theory

**Lemma 0.0.1.** Suppose that  $\varphi : F \rightarrow F'$  is a Field-Homomorphism. If  $\varphi$  is not zero map, then  $\varphi$  is injective.

*Proof.* Since  $\varphi$  is not zero map,  $\varphi(a) \neq 0_{F'}$  for some  $a \in F$ . Thus,  $\ker \varphi \subsetneq F$ .

But, the field  $F$  has only ideal for  $\{0\}$  or  $F$ , hence the ideal  $\ker \varphi$  must be  $\{0\}$ . Now,

$$\varphi(a) = \varphi(b) \implies \varphi(a - b) = 0 \implies a - b = 0$$

□

**Lemma 0.0.2.** Suppose that  $F$  is a field.

If  $p(x) \in F[x]$  is irreducible in  $F[x]$ , then  $F[x]/(p(x))$  is a field.

*Proof.* Since  $F$  is a field,  $F[x]$  is Euclidean domain, P.I.D., and U.F.D. Thus,

$$p(x) \text{ irreducible} \xRightarrow{\text{U.F.D.}} (p(x)) \text{ prime} \xRightarrow{\text{P.I.D.}} (p(x)) \text{ maximal in } F[x]$$

Hence,  $K = F[x]/(p(x))$  becomes a field.

□

**Theorem 0.0.3.** Suppose that  $F$  is a field, and  $p(x) \in F[x]$  is irreducible.

Then,  $K \stackrel{\text{def}}{=} F[x]/(p(x))$  is a field containing an isomorphic copy of  $F$ , and  $p(x)$  has a root  $\alpha$  in  $K$ .

*Proof.* Since  $F$  is a field,  $F[x]$  is Euclidean domain, P.I.D., and U.F.D. Thus,

$$p(x) \text{ irreducible} \xRightarrow{\text{U.F.D.}} (p(x)) \text{ prime} \xRightarrow{\text{P.I.D.}} (p(x)) \text{ maximal in } F[x]$$

Hence,  $K = F[x]/(p(x))$  becomes a field. Consider the Canonical projection

$$\pi : F[x] \rightarrow F[x]/(p(x)) : f(x) \mapsto f(x) + (p(x))$$

and restriction over  $F$ ,  $\varphi = \pi_F : F \rightarrow F[x]/(p(x))$  is Homomorphism with  $\ker \varphi = \{0\}$ , because  $\varphi(1) + (p(x)) \neq (p(x))$ . Now,  $F \cong F/\ker \varphi \cong \varphi[F] \subseteq K = F[x]/(p(x))$ . Further, denote  $\bar{x} \stackrel{\text{def}}{=} \pi(x) = x + (p(x))$ , and  $p(x) = \sum_{i=0}^n a_i x^i$ . Then,

$$p(\bar{x}) = \sum_{i=0}^n \overline{a_i x^i} = \sum_{i=0}^n \pi(a_i) \pi(x)^i = \pi \left( \sum_{i=0}^n a_i x^i \right) = \pi(p(x)) = \overline{p(x)} = \bar{0} \in K$$

□

**Theorem 0.0.4.** Suppose that  $F$  is a field, and  $p(x) \in F[x]$  is irreducible. If  $K$  is an Extension field of  $F$  containing a root  $\alpha \in K$  of  $p(x)$ , then  $F(\alpha) \cong F[x]/(p(x))$ .

*Proof.* Define a map  $\varphi_\alpha : F[x]/(p(x)) \rightarrow F(\alpha) : f(x) + (p(x)) \mapsto f(\alpha)$ .

**Well-defined:** Suppose that  $f(x) + (p(x)) = g(x) + (p(x))$ . Then,  $\varphi_\alpha(f(x) + (p(x))) = f(\alpha)$  and  $\varphi_\alpha(g(x) + (p(x))) = g(\alpha)$ .

$$f(x) + (p(x)) = g(x) + (p(x)) \iff f(x) - g(x) \in (p(x))$$

$$f(x) - g(x) \in (p(x)) \implies f(x) - g(x) = p(x)r(x) \text{ for some } r(x) \in F[x] \implies f(\alpha) - g(\alpha) = p(\alpha)r(\alpha) = 0 \cdot r(\alpha) = 0.$$

**Ring-Homomorphism:**

1.  $\varphi([f(x) + (p(x))] + [g(x) + (p(x))]) = \varphi(f(x) + g(x) + (p(x))) = f(\alpha) + g(\alpha) = \varphi(f(x) + (p(x))) + \varphi(g(x) + (p(x)))$ .
2.  $\varphi([f(x) + (p(x))] \cdot [g(x) + (p(x))]) = \varphi(f(x)g(x) + (p(x))) = f(\alpha)g(\alpha) = \varphi(f(x) + (p(x))) \cdot \varphi(g(x) + (p(x)))$ .

**Injectivity:** Since  $\varphi$  is a non-trivial homomorphism from field into field, thus  $\ker \varphi = \{0\}$ .

(More precisely,  $\varphi(1) = 1 + (p(x)) \neq (p(x))$  because  $1 \notin (p(x))$ . Thus,  $\ker \varphi \subsetneq F[x]/(p(x)) \implies \ker \varphi = \{0\}$ .)

Now, the  $\varphi$  becomes an Embedding:

$$\varphi[F[x]/(p(x))] \cong F[x]/(p(x))$$

Since  $F[x]/(p(x))$  is a field, the image  $\varphi[F[x]/(p(x))]$  is a field, being it is isomorphic copy.

The definition of  $F(\alpha)$  gives  $\varphi[F[x]/(p(x))] \supset F(\alpha)$ , and  $\varphi[F[x]/(p(x))] \subset F(\alpha)$  is trivial. Consequently,

$$F(\alpha) = \varphi[F[x]/(p(x))] \cong F[x]/(p(x))$$

□

**Theorem 0.0.5.** Suppose that  $p(x) \in F[x]$  is an irreducible with  $\deg p(x) = n$ , and  $K = F[x]/(p(x))$ . Put  $\bar{x} = x + (p(x)) \in K$ . Then,  $\{1, \bar{x}, \bar{x}^2, \dots, \bar{x}^{n-1}\}$  is basis for  $K$  as Vector space over  $F$ .

*Proof.* Above lemma guarantees  $K = F[x]/(p(x))$  is a field.

1) *Linealy independent.*

Suppose that  $a_0 + a_1\bar{x} + \dots + a_{n-1}\bar{x}^{n-1} = 0$  in  $K$  where  $a_i$  are not all zero. Then,

$$\begin{aligned} 0 &= a_0 + a_1\bar{x} + \dots + a_{n-1}\bar{x}^{n-1} \\ &= a_0 + a_1(x + (p(x))) + \dots + a_{n-1}(x^{n-1} + (p(x))) \\ &= a_0 + a_1x + \dots + a_{n-1}x^{n-1} + (p(x)) \\ &\iff a_0 + a_1x + \dots + a_{n-1}x^{n-1} = p(x)b(x) \text{ for some } b(x) \in F[x] \end{aligned}$$

But, the left term has a degree at most  $n-1$ , and the right term has a degree at least  $n$ . Contradiction.

2)  $F[x]$  is generated by  $\{1, \bar{x}, \dots, \bar{x}^{n-1}\}$ .

Let  $a(x) \in F[x]$ . Then, the Euclidean algorithm for  $F[x]$  gives that: there exist uniquely  $q(x), r(x) \in F[x]$  s.t

$$a(x) = q(x)p(x) + r(x) \text{ with } \deg r(x) < \deg p(x)$$

Since  $q(x)p(x) \in (p(x))$ ,  $a(x) + (p(x)) = r(x) + (p(x))$ , the degree of  $a(x) + (p(x))$  is smaller than  $n$ .

Now, for any  $a(x) + (p(x)) \in K = F[x]/(p(x))$ ,  $a(x) + (p(x)) \in \langle 1, \bar{x}, \dots, \bar{x}^{n-1} \rangle$ .

□

**Corollary 0.0.6.** Suppose that  $F$  is a field, and  $p(x) \in F[x]$  is irreducible polynomial with  $\deg p(x) = n$ . If  $E$  is an Extension field of  $F$  containing  $\alpha$  as a root of  $p(x)$ . Then,

$$F(\alpha) = \{a_0 + a_1\alpha + a_2\alpha^2 + \dots + a_{n-1}\alpha^{n-1} \mid a_i \in F\} \subseteq E$$

*Proof.* Since  $F[x]/(p(x)) = \{a_0 + a_1\bar{x} + \dots + a_{n-1}\bar{x}^{n-1} \mid a_i \in F\} \cong F(\alpha)$ , being the isomorphsim

$$\varphi_\alpha : F[x]/(p(x)) \rightarrow F(\alpha) : f(x) + (p(x)) = \overline{f(x)} \mapsto f(\alpha)$$

Thus, combining with above fact,

$$\begin{aligned} F(\alpha) &= \varphi_\alpha[F[x]/(p(x))] = \{\varphi_\alpha(a_0 + a_1\bar{x} + \dots + a_{n-1}\bar{x}^{n-1}) \mid a_i \in F\} \\ &= \{\varphi_\alpha(a_0) + \varphi_\alpha(a_1\bar{x}) + \dots + \varphi_\alpha(a_{n-1}\bar{x}^{n-1}) \mid a_i \in F\} \\ &= \{a_0 + a_1\alpha + a_2\alpha^2 + \dots + a_{n-1}\alpha^{n-1} \mid a_i \in F\} \end{aligned}$$

□

## 0.0.2 Extension Field

**Definition 0.0.7.** Suppose that  $F$  is a field, and  $K$  be a field containing  $F$  as a subfield. The element  $\alpha \in K$  is called *algebraic* over  $F$  if:

There exists a non-zero polynomial  $f(x) \in F[x]$  such that  $f(\alpha) = 0$  in  $K$ .

If not algebraic, it is called *transcendental* over  $F$ .

**Theorem 0.0.8.** Suppose that  $\alpha \in K$  is algebraic over  $F$ .

Then, there exists a unique monic irreducible polynomial  $m_{\alpha,F}(x) \in F[x]$  such that  $m_{\alpha,F}(\alpha) = 0$ . Moreover,

$f(x) \in F[x]$  has  $\alpha$  as a root if and only if  $m_{\alpha,F}(x) \in F[x]$  divides  $f(x)$  in  $F[x]$ .

*Proof.* Since  $\alpha \in K$  is algebraic over  $F$ , the Well-Ordering principle gives the existence of minimal degree polynomial  $g(x) \in F[x]$  having  $\alpha$  as a root. More precisely,

$$S = \{\deg f(x) \in \mathbb{N} \mid f(x) \in F[x] \text{ s.t. } f(\alpha) = 0\}$$

is non-empty since  $\alpha$  is algebraic. Therefore, the Well-Ordering Principle gives that  $S$  has a minimum  $n \in \mathbb{N}$ .

Now, we can choose  $g(x) \in F[x]$  with  $\deg g(x) = n$  and  $g(\alpha) = 0$ .

Using Contradiction: Suppose that  $g(x)$  is reducible. That is,  $g(x) = a(x)b(x)$  with  $\deg a, \deg b < \deg g$ .

Since  $g(\alpha) = a(\alpha)b(\alpha) = 0$  and  $K$  is a field, either  $a(\alpha)$  or  $b(\alpha)$  must be zero.

This contradicts with  $n \in \mathbb{N}$  is the minimum degree of polynomial which has  $\alpha$  as a root.

In summary:

If  $g(x) \in F[x]$  is a polynomial of minimal degree which has  $\alpha$  as a root, then  $g(x)$  is irreducible over  $F$ .

Now, Suppose that  $f(x) \in F[x]$  is any polynomial having  $\alpha$  as a root. Then,  $\deg f(x) \geq \deg g(x)$  by setting.

By the Euclidean Algorithm in  $F[x]$ , there exist unique polynomials  $q(x), r(x) \in F[x]$  such that

$$f(x) = q(x)g(x) + r(x) \text{ with } \deg r(x) < \deg g(x)$$

Meanwhile,

$$0 = f(\alpha) = q(\alpha)g(\alpha) + r(\alpha) = q(\alpha) \cdot 0 + r(\alpha) = r(\alpha)$$

$r(\alpha) = 0$  implies  $r(x)$  must be zero, by minimality. Hence,  $g(x)$  divides  $f(x)$  in  $F[x]$ .

Finally, put  $a \in F$  be a leading coefficient of  $g(x)$ . Then,  $m_{\alpha,F}(x) = a^{-1}g(x)$  is monic polynomial.

And, Uniqueness given by: for any  $f(x) \in F[x]$ ,  $f(\alpha) = 0 \implies m_{\alpha,F}(x) \mid f(x)$ . □

**Corollary 0.0.9.** If  $L$  is an Extension of field  $F$  and  $\alpha \in L$  is algebraic over both  $F$  and  $L$ , then  $m_{\alpha,L}(x)$  divides  $m_{\alpha,F}(x)$  in  $L[x]$ .

Above theorem allows defining:

**Definition 0.0.10.** Suppose that  $K$  is an Extension field of a field  $F$ , and  $\alpha \in K$  is algebraic over  $F$ . The monic irreducible polynomial which has  $\alpha$  as root is called the *minimal polynomial* for  $\alpha$  over  $F$ . Denote this  $m_{\alpha,F}(x) \in F[x]$  or  $\text{irr}(\alpha, F)$ .

**Proposition 0.0.11.** Suppose that  $F$  is a field and  $E$  is an Extension of  $F$ .

If  $\alpha \in E$  is algebraic over  $F$ , then  $F(\alpha) \cong F[x]/(m_{\alpha,F}(x))$  and

$$[F(\alpha) : F] \stackrel{\text{def}}{=} \dim_F F(\alpha) = \deg m_{\alpha,F}(x) \stackrel{\text{def}}{=} \deg \alpha$$

**Theorem 0.0.12.** Suppose that  $E$  is an Extension field of a field  $F$ , and  $\alpha \in E$ .

$\alpha \in E$  is algebraic over  $F$  if and only if  $[F(\alpha) : F]$  has finite dimension

*Proof.* If  $\alpha \in E$  is algebraic over  $F$ , then  $[F(\alpha) : F] = \deg m_{\alpha, F}(x)$ , thus finite.

Precisely, if  $f(x) \in F[x]$  satisfies  $\deg f(x) = n$  and  $f(\alpha) = 0$ , then  $[F(\alpha) : F] = \deg_F \alpha \leq n$ .

Conversely, suppose  $\alpha \in E$  where  $[F(\alpha) : F] = n$ . Then, the  $n + 1$  elements

$$1, \alpha, \alpha^2, \dots, \alpha^n$$

must be linearly dependent, thus for some not all zero elements  $b_0, \dots, b_n \in F$ ,

$$b_0 + b_1\alpha + b_2\alpha^2 + \dots + b_n\alpha^n = 0$$

Now, the polynomial in  $F[x]$

$$f(x) = b_0 + b_1x + b_2x^2 + \dots + b_nx^n$$

satisfies  $f(\alpha) = 0$  in  $F(\alpha)$ . In summary, if  $[F(\alpha) : F] = n$ , then  $\alpha \in E$  is algebraic over  $F$  with  $\deg_F \alpha \leq n$ .  $\square$

**Corollary 0.0.13.** If  $E$  is an Extension field of  $F$  with  $[E : F]$  has finite, then  $E$  is algebraic over  $F$ .

**Theorem 0.0.14.** Suppose that  $F \subseteq L \subseteq E$  are fields. Then,

$$[E : F] = [E : L][L : F]$$

*Proof.* Suppose that  $[E : L]$  and  $[L : F]$  are finite. Put  $[E : L] = m$  and  $[L : F] = n$ .

Set  $A = \{\alpha_1, \alpha_2, \dots, \alpha_m\} \subset E$  and  $B = \{\beta_1, \beta_2, \dots, \beta_n\} \subset L$  are basis of  $E$  over  $L$  and  $L$  over  $F$ , respectively.

Let  $x \in E$  be given. Since  $A$  is basis for  $E$  over  $L$ , there exists a unique linear combination: for  $a_i \in L$ ,

$$x = a_1\alpha_1 + a_2\alpha_2 + \dots + a_m\alpha_m = \sum_{i=1}^m a_i\alpha_i$$

Meanwhile, for each  $a_i \in L$ ,  $1 \leq i \leq m$ , there exist unique linear combinations: for  $b_{i,j} \in F$ ,

$$a_i = b_{i,1}\beta_1 + b_{i,2}\beta_2 + \dots + b_{i,n}\beta_n = \sum_{j=1}^n b_{i,j}\beta_j$$

Combining above, we obtain

$$x = \sum_{i=1}^m \left( \sum_{j=1}^n b_{i,j}\beta_j \right) \alpha_i = \sum_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n}} b_{i,j}\beta_j\alpha_i$$

Since each  $b_{i,j} \in F$  and  $\beta_j\alpha_i \in E$ , the subset  $C = \{\beta_j\alpha_i \mid \alpha_i \in A, \beta_j \in B\} \subset E$  spans  $E$  over  $F$ .

Meanwhile, this  $C$  is linearly independent over  $F$ , because: Suppose that

$$\sum_{i=1}^m \left( \sum_{j=1}^n c_{i,j}\beta_j \right) \alpha_i = 0$$

Since each  $c_{i,j}\beta_j \in L$  and  $A = \{\alpha_i\}$  is linearly independent over  $L$ , for each  $1 \leq i \leq m$ ,

$$\sum_{j=1}^n c_{i,j}\beta_j = 0$$

$c_{i,j} \in F$  and  $B = \{\beta_j\}$  is linearly independent over  $F$ , all  $c_{i,j} = 0$ . Thus,  $C$  is basis which has  $mn$  elements.

Meanwhile, if  $[L : F]$  is infinite, then  $[E : F]$  is clearly infinite; being  $E \subset L$ . Similarly,  $[E : L]$ .  $\square$

**Theorem 0.0.15.** Suppose that  $E$  is an algebraic Extension field of  $F$ . Then,

$$E \text{ is finite Extension} \iff \text{There exist finite elements } \alpha_1, \dots, \alpha_n \in E \text{ such that } E = F(\alpha_1, \dots, \alpha_n).$$

*Proof.* Suppose that  $E$  is finite Extension.

If  $[E : F] = 1$ , then  $F(1) = F = E$ , claim proved.

If  $[E : F] > 1$ , then  $E \neq F$ . Thus, put  $\alpha_1 \in E \setminus F$ . If  $F(\alpha_1) = E$ , then there is nothing to prove. If  $F(\alpha_1) \neq E$ , then put again  $\alpha_2 \in E \setminus F(\alpha_1)$ . Since  $[E : F]$  is finite integer, this process will terminate in finite.

Thus for some  $n \in \mathbb{N}$ ,  $E = F(\alpha_1, \dots, \alpha_n)$ .

Conversely, suppose that  $E = F(\alpha_1, \dots, \alpha_n)$  for some  $\alpha_i \in E$ . Then,

$$E = [E : F(\alpha_n)][F(\alpha_n) : F(\alpha_{n-1})] \cdots [F(\alpha_1) : F]$$

Since  $E$  is algebraic,  $[E : F(\alpha_n)], [F(\alpha_n) : F(\alpha_{n-1})], \dots, [F(\alpha_1) : F]$  are all finite. Thus  $[E : F]$  is finite. □

**Theorem 0.0.16.** If  $E$  is Algebraic over  $K$  and  $K$  is Algebraic over  $F$ . Then,  $E$  is Algebraic over  $F$ .

*Proof.* Let  $\alpha \in E$  be given. Then, there exist not all zero  $a_0, \dots, a_n \in K$  such that

$$a_n \alpha^n + a_{n-1} \alpha^{n-1} + \cdots + a_1 \alpha + a_0 = 0$$

Since each  $a_i \in K$  is algebraic over  $F$ , thus  $F(\alpha_0, \dots, \alpha_n)$  is finite over  $F$ , being above theorem. Further,

$$F(\alpha, a_0, \dots, a_n) = F(a_0, \dots, a_n)(\alpha)$$

is finite over  $F(a_0, \dots, a_n)$ , because above equation gives that  $\alpha$  is algebraic over  $F(a_0, \dots, a_n)$ . Now,

$$[F(\alpha, a_0, \dots, a_n) : F] = [F(\alpha, a_0, \dots, a_n) : F(a_0, \dots, a_n)][F(a_0, \dots, a_n) : F]$$

is finite, thus algebraic over  $F$ . This means  $\alpha \in E$  is algebraic over  $F$ . □

**Theorem 0.0.17.** Let  $E$  be an Extension field of  $F$ , and  $\alpha \in E$ .

$$F(\alpha) = \left\{ \frac{p(\alpha)}{q(\alpha)} \mid p(x), q(x) \in F[x], q(\alpha) \neq 0 \right\}$$

### 0.0.3 Algebraic Closure

**Lemma 0.0.18.** Suppose that  $E$  is an Extension field of  $F$ . Then,

$$\overline{F}_E \stackrel{\text{def}}{=} \{\alpha \in E \mid \alpha \text{ is algebraic over } F\}$$

is a subfield of  $E$ . This  $\overline{F}_E$  is called *Algebraic Closure* of  $F$  in  $E$ .

*Proof.* Let  $\alpha, \beta \in \overline{F}_E$  be given. Since  $\alpha$  is algebraic over  $F$ , thus  $F(\alpha)$  is finite extension of  $F$ . Then,  $F(\alpha, \beta) = F(\alpha)(\beta)$  is finite over  $F(\alpha)$ , thus  $F(\alpha, \beta)$  is finite extension over  $F$ , thus algebraic over  $F$ . This implies  $F(\alpha, \beta) \subseteq \overline{F}_E$ . Now, with assumption  $\beta \neq 0$ ,

$$\alpha + \beta, \alpha\beta, \frac{\alpha}{\beta} \in F(\alpha, \beta) \subseteq \overline{F}_E$$

□

**Definition 0.0.19.** A field  $F$  is said to be *Algebraically Closed* if:

Every non-constant polynomial in  $F[x]$  has a zero in  $F$

**Theorem 0.0.20.** A field  $F$  is Algebraically Closed if and only if

every non-constant polynomial in  $F[x]$  factors into linear factors within  $F[x]$ .

*Proof.* Suppose that  $F$  is Algebraically Closed.

Let  $f(x) \in F[x]$  be given. Since  $f(x)$  has a root in  $F$ , put  $\alpha_0 \in F$  such that  $f(\alpha_0) = 0$ . Then,

$$f(x) = (x - \alpha_0)f_1(x)$$

And,  $f_1(x) \in F[x]$ , it has a root in  $F$ , put  $\alpha_1 \in F$  such that  $f_1(\alpha_1) = 0$ . Now,

$$f(x) = (x - \alpha_0)(x - \alpha_1)f_2(x)$$

In finite process, we get:

$$f(x) = c(x - \alpha_0)(x - \alpha_1) \cdots (x - \alpha_{\deg f(x)})$$

Conversely argument is clear.

□

**Lemma 0.0.21.** Suppose that  $F$  is a field, and  $f(x) \in F[x]$  is non-constant polynomial.

Then, there exists an Extension  $E$  such that  $f(x)$  has a root in  $E$ .

*Proof.* Let  $f(x) \in F[x]$ . Since  $F[x]$  is U.F.D., factors  $f(x)$  into irreducibles in  $F[x]$ ,

$$f(x) = f_1(x) \cdots f_n(x)$$

Now, for any  $1 \leq j \leq n$ ,  $F[x]/(f_j(x))$  is an Extension of  $F$  containing a root of  $f(x)$ .

□

### 0.0.4 † Existence of Algebraic Closure

**Definition 0.0.22.** Suppose that  $F$  is a field. An Extension  $\overline{F}$  is called *Algebraic Closure* of  $F$  if:

1.  $\overline{F}$  is Algebraic over  $F$ .
2. Every nonconstant polynomial  $f(x) \in F[x]$  factors completely into linear factors over  $\overline{F}$ .

**Theorem 0.0.23.** An Algebraic Closure  $\overline{F}$  is Algebraically Closed.

*Proof.* Let  $f(x) \in \overline{F}[x]$  be given. Then, there exists an Extension  $E$  of  $\overline{F}$  containing a root  $\alpha \in E$  of  $f(x)$ . Now,  $\overline{F}(\alpha) \subset E$  is algebraic over  $\overline{F}$ , and by definition,  $\overline{F}$  is algebraic over  $F$ . Thus  $\overline{F}(\alpha)$  is algebraic over  $F$ . Particulary,  $\alpha$  is algebraic over  $F$ , thus  $\alpha \in \overline{F}$ . □

**Theorem 0.0.24.** For any field  $F$ , there exists an Algebraically closed field  $K$  containing  $F$ .

*Proof.* Denote the set  $M \stackrel{\text{def}}{=} \{f \in F[x] \mid \deg f \geq 1\}$ . Set  $S \stackrel{\text{def}}{=} \{x_f \mid f \in M\}$ . Define the Polynomial Ring

$$F[S] \stackrel{\text{def}}{=} \bigcup_{\substack{\mathcal{F} \subset S \\ \mathcal{F} \text{ finite}}} F[\mathcal{F}]$$

Consider the ideal

$$I \stackrel{\text{def}}{=} (\{f(x_f) \mid f \in M\}) \subseteq F[S]$$

If  $I = F[S]$ , that is,  $I$  is Entire, then  $1 \in I$ . Thus, for some  $g_1, \dots, g_n \in F[S]$  such that

$$g_1 f_1(x_{f_1}) + g_2 f_1(x_{f_2}) + \dots + g_n f_n(x_{f_n}) = 1$$

For simplicity, write  $x_i = x_{f_i}$ . Since each  $g_i$  has only a finite number of variables, for  $m \geq n$ , write

$$\sum_{i=1}^n g_i(x_1, \dots, x_n, \dots, x_m) f_i(x_i) = 1$$

For each  $1 \leq i \leq n$ , let  $F_i$  be an Extension of  $F$  such that  $f_i$  has a root  $\alpha_i$ .

Then, in the Composition Field  $F_1 \cdots F_n$ ,

$$1 = \sum_{i=1}^n g_i(x_1, \dots, x_n, \dots, x_m) f_i(\alpha_i) = 0$$

But this is Contradiction, thus  $I$  is Proper Ideal. The Zorn's Lemma gives the Existence of a Maximal Ideal

$$I \subsetneq \mathcal{M} \subsetneq F[S]$$

Thus, the Quotient ring

$$K_1 \stackrel{\text{def}}{=} F[S]/\mathcal{M}$$

is a Field containing  $F$ , and each polynomial  $f$  has a root in  $K_1$ .

Inductively, construct  $K_2$  as an Extension of  $K_1$  containing a root of all polynomial in  $K_1$ . We obtain

$$F = K_0 \subseteq K_1 \subseteq K_2 \subseteq \dots \subseteq K_j \subseteq \dots$$

Finally, define

$$K = \bigcup_{j=0}^{\infty} K_j$$

Then,  $K$  contains  $F$ , and any polynomial  $k(x) \in K[x]$ , for some  $j \geq 0$ ,  $k(x) \in K_j[x]$ . And,  $k(x)$  has a root in  $K_{j+1}[x]$  by construction, thus  $K$  is Algebraically Closed. □

**Theorem 0.0.25.** Suppose that  $E$  is an Extension field of  $F$ , and  $Q$  is a Field of Fractions of  $F[x]$ . Then,  
 $E$  is Not Algebraic over  $F$  if and only if There exists an Embedding  $\varphi : Q \rightarrow E$  such that  $\forall a \in F, \varphi(a) = a$ .